

FUSIONCORE v_0

Executive Technical Report

Phases 1–5 Programme Summary

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1. Abstract

FusionCore v_0 is a physics-aware predictive maintenance pipeline for turbofan engine Remaining Useful Life (RUL) estimation, developed against the NASA C-MAPSS benchmark dataset comprising 709 training engines and 707 official test engines across four operating subsets (FD001–FD004). The programme spans five sequential phases: physics-grounded exploratory analysis (Phase 1), zero-leakage preprocessing and dataset unification (Phase 2), physics-aware feature engineering (Phase 3), SOTA benchmarking and forensic audit (Phase 4), and Iron Wall Protocol evaluation against the official test set (Phase 5).

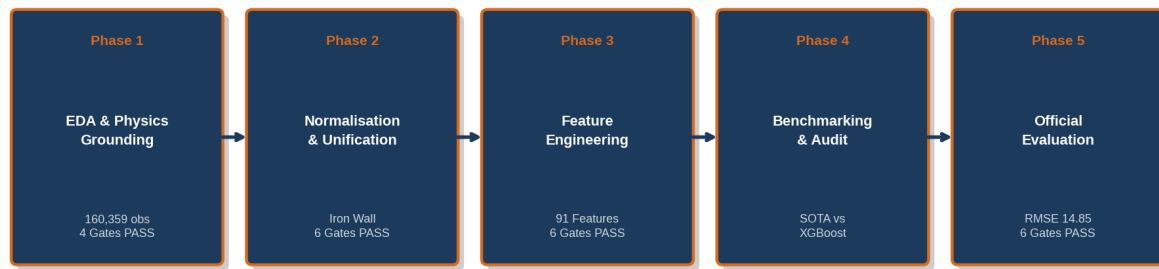
The XGBoost baseline, trained on 91 engineered features, achieves an RMSE of 14.85 cycles, MAE of 10.34 cycles, and a NASA asymmetric score of 4,336.3 on the 707-engine official test set. All 22 programmatic gate conditions across the five phases pass without exception. At fleet scale, these results translate to an estimated annual net saving of approximately \$2.5M for a 100-aircraft narrowbody operator at baseline financial parameters.

FusionCore v_0 is complete and cleared for transition to FusionCore v_1 , which will introduce PiNet — a hybrid physics-informed neural network architecture designed to address the temporal degradation modelling limitations identified during Phase 4 benchmarking.

2. Programme Overview

FusionCore v_0 addresses a well-characterised gap in aerospace predictive maintenance: the absence of a physics-grounded, zero-leakage RUL estimation pipeline that is simultaneously rigorous, interpretable, and traceable to published aerospace engineering literature. The commercial motivation is direct. Unscheduled engine removals (UERs) are the single largest driver of unplanned maintenance cost and aircraft-on-ground (AOG) events in commercial aviation. A narrowbody AOG event incurs direct costs in the range \$50,000–\$150,000 per day. Accurate RUL estimation — particularly discrimination between engines in the Critical operational band (RUL 0–30 cycles) and those in the Healthy band (RUL 60–125 cycles) — enables condition-based maintenance scheduling that directly reduces both over-maintenance and unplanned failure.

The programme is structured as a sequential five-phase pipeline in which each phase produces verified outputs that form the mandatory inputs to the next. No phase bypasses its predecessor. All analytical claims are verified by programmatic gate conditions that halt execution on failure, preventing uncertified outputs from propagating through the pipeline. The overall architecture is illustrated in Figure 1.

Figure 1 — FusionCore v_0 Five-Phase Pipeline Architecture

The dataset underpinning the entire programme is the NASA C-MAPSS (Commercial Modular Aero-Propulsion System Simulation) turbofan degradation dataset (Saxena & Goebel, 2008), which provides four simulation subsets encoding single-fault and dual-fault degradation modes under one and six operating regimes respectively. All 709 training engines are used for model development; the 707 official test engines are held back under the Iron Wall Protocol until Phase 5 final evaluation.

3. Phase 1 — Exploratory Data Analysis & Physics Grounding

Phase 1 establishes the empirical and physical foundations of the FusionCore pipeline. Its primary objective is to verify that the C-MAPSS corpus encodes genuine turbofan degradation physics before any preprocessing or modelling decisions are committed. This verification is non-negotiable: a pipeline built on data that does not reflect real physics produces spurious correlations that cannot be trusted in operational deployment.

3.1 Key Findings

The corpus contains 160,359 engine-cycle observations across 709 training engines. Lifespan analysis reveals substantial variability: engine lifespans range from 128 to 543 cycles across subsets, with the dual-fault, six-regime subset FD004 exhibiting the highest complexity. This variability is itself the commercial case for predictive maintenance — a fixed time-based maintenance schedule cannot simultaneously protect against AOG risk and avoid chronic over-maintenance across a population with this spread.

Reference: Phase 1 Technical Analysis, Section 2.2

Variance analysis across all 21 sensors identifies seven dead sensors (s1, s5, s6, s10, s16, s18, s19) in the single-regime subsets FD001 and FD003. These sensors carry no degradation information in narrow operational profiles and their exclusion from kinematic expansion in Phase 3 is physically justified, not arbitrary. Autocorrelation analysis confirms that turbofan health state is a memory process: the ACF and PACF of the three primary degradation sensors (s4 EGT, s7 P30, s9 Nc) exhibit significant serial correlation with decay constants consistent with progressive mechanical degradation.

3.2 Gate Outcomes

Phase 1 Gate Verification: PASS — 4 gates

Gate 1 (Physical Plausibility): FD001 s1 reports a mean of 518.67 °R, matching the ISA standard atmosphere value. **Gate 2** (Dead Sensor Identification): seven dead sensors recovered exactly against the Saxena & Goebel (2008) canonical list. **Gate 3** (KDE Classification): dead sensors produce near-

singular distributions; active sensors produce continuous degradation-correlated spread. Gate 4 (Lifecycle Trajectories): degradation trends confirmed visually across all four subsets.

Reference: Phase 1 Technical Analysis, Sections 3–5.

4. Phase 2 — Zero-Leakage Normalisation & Dataset Unification

Phase 2 constructs the preprocessing infrastructure that transforms raw C-MAPSS sensor readings into a clean, regime-normalised, leakage-free feature space ready for model training. The central engineering concern at this stage is data leakage: any normalisation parameter computed using validation or test data produces an optimistic bias that invalidates evaluation results. Phase 2 eliminates this risk through the Iron Wall Protocol.

4.1 Key Deliverables

Iron Wall Protocol (`GroupShuffleSplit`): Partitioning is performed at engine level using `GroupShuffleSplit`, ensuring that no single engine contributes cycles to both the training and validation splits. The resulting partition achieves an 80/20 engine-level split across all four subsets. This is the foundational leakage guard for the entire programme (see).

Reference: Phase 2 Technical Analysis, Section 3

Operating Regime Identification: K-Means clustering with $K = 6$ is applied to the three operational settings (op1, op2, op3) for the multi-regime subsets FD002 and FD004. The six cluster centroids correspond exactly to the six discrete operating conditions in the C-MAPSS simulation. The K-Means models are serialised and frozen for consistent application to validation and test data.

Per-Regime Z-Score Normalisation: Normalisation parameters (μ, σ) are fitted exclusively on the internal training split per regime, then applied frozen to validation and test data. A near-zero sigma guard (variance threshold $\varepsilon = 10^{-4}$) prevents division by zero for dead sensors. All 21 sensor channels are normalised; regime-dead sensors produce near-zero variance post-normalisation and are handled explicitly in Phase 3.

P-F Interval Analysis: Shewhart Statistical Process Control analysis quantifies the P-F interval — the schedulable window between first alert and functional failure. The overall P-F interval coefficient of variation (CV) of 1.052 exceeds the engineering threshold of 0.50, indicating wide spread. This advisory finding directly drives the survival analysis work in Phase 3 and the family-specific threshold recalibration.

Reference: Phase 2 Technical Analysis, Section 5

4.2 Gate Outcomes

Phase 2 Gate Verification: PASS — 6 gates, P-F CV advisory noted

Reference: Phase 2 Technical Analysis, Sections 3–6.

5. Phase 3 — Physics-Aware Feature Engineering

Phase 3 serves a dual mandate: systematic resolution of every outstanding gap from Phases 1 and 2, and construction of the 91-feature physics-aware matrix that feeds all models in Phase 4. The feature engineering is grounded in published turbofan thermodynamics and degradation physics — each feature is physically motivated, not empirically discovered.

5.1 Gap Resolution

Thirteen gaps were carried forward from Phases 1 and 2. Seven were resolved directly in Phase 3 (mandatory items); the remaining six were documented with justified deferrals and transferred as transfer items to Phase 4. Resolutions include composite key enforcement for engine uniqueness, one-hot encoding of the nominal regime identifier, VIF analysis and ratio feature construction for mechanically coupled sensor pairs (s8/s13, s9/s14), extreme Z-score excursion investigation, and the Out-of-Distribution guard for regime assignment at deployment.

Reference: Phase 3 Technical Analysis, Section 4

5.2 Feature Matrix Composition

Table 1 — Phase 3 Feature Matrix Composition (91 Features)

Category	Count	Description
Core manifold	24	3 operational settings + 21 z-normalised sensors
Kinematic expansion	51	delta, rolling mean, rolling std on 17 active sensors
Virtual sensors	3	CPR, E_{thermal} , EGT_{drift}
Fatigue indices	3	Miner's Rule cumulative damage for s4, s7, s9
Gap resolution features	10	Ratio features, one-hot regime, health index
Total	91	All features verified by programmatic count assertion

Virtual sensor formulae:

$$CPR = \frac{s7}{s3} \text{ (Compressor Pressure Ratio)}$$

$$E_{\text{thermal}} = \frac{s12 \times s11}{s9} \text{ (Thermal Efficiency Proxy)}$$

$$EGT_{\text{drift}} = z_{s4} - \mu_{\text{baseline}} \text{ (EGT margin)}$$

All three encode domain knowledge about the turbofan thermodynamic cycle that the raw sensor channels do not represent directly.

5.3 Survival Analysis

A Kaplan-Meier and Cox Proportional Hazards (Cox, 1972) survival analysis framework is applied to the P-F interval distribution. The Cox model identifies EGT deviation (s4) and corrected core speed (s9) as the two dominant hazard predictors. The log-rank test (Mantel, 1966) confirms that single-fault and dual-fault engine families exhibit statistically distinct survival profiles, motivating the family-specific UWL threshold recalibration that reduces the P-F interval spread advisory from Phase 2.

Reference: Phase 3 Technical Analysis, Section 4.12

5.4 Gate Outcomes

Phase 3 Gate Verification: PASS — 6 gates

Reference: Phase 3 Technical Analysis, Sections 4–8.

6. Phases 4 & 5 — Benchmarking, Forensic Audit & Official Evaluation

Phases 4 and 5 constitute the model training, validation, and final evaluation stages of FusionCore v_0 . Phase 4 trains and audits all candidate models on the internal train/validation split. Phase 5 applies the Iron Wall Protocol to the official NASA test set — models are evaluated once, with no retraining permitted after results are observed.

6.1 Transfer Item Resolution

Nine transfer items from Phase 3 were resolved prior to model training. All mandatory items (T1–T9) were resolved programmatically. Key resolutions include: Yeo-Johnson skewness correction for CPR and E_{thermal} (applied to neural network inputs only; XGBoost is invariant to monotone transforms); inverse-frequency regime weighting for neural network training; CPR denominator guard (s3 degeneracy); and `fillna(0)` convention for trajectory boundary features.

Reference: Phase 4/5 Technical Analysis, Appendix D

6.2 XGBoost Baseline

XGBoost (Chen & Guestrin, 2016) was trained on the internal training split (129,331 rows $\times$ 91 features) with early stopping against the validation split (31,028 rows). SHAP (Lundberg & Lee, 2017) TreeExplainer analysis on a 5,000-row validation sample confirms that the top 20 features by mean absolute SHAP value account for 88.5% of total model attribution, with kinematic delta and rolling-mean features of s4, s7, and s9 dominating — consistent with the degradation physics established in Phase 1.

Reference: Phase 4/5 Technical Analysis, Section 3.3

Forensic audit: A target null test (random permutation of the training target) reduces performance to noise levels, confirming genuine feature–target correlation rather than row-index memorisation. A perturbation reactivity test confirms physical responsiveness: a synthetic s7-family fault injected into a healthy engine produces the expected monotonic RUL reduction.

Reference: Phase 4/5 Technical Analysis, Section 3.4

6.3 SOTA Benchmarks

DeepAR: Disqualified on two programmatically confirmed architectural grounds: inability to ingest exogenous sensor features, and mandatory look-ahead contamination. No model training was performed.

Reference: Phase 4/5 Technical Analysis, Section 3.5

TFT and N-HITS: Both architectures were trained under capacity-scaled hyperparameters with `exclude_insample_y=True` enforced. TFT and N-HITS produce identical performance metrics to four decimal places across both the validation and test sets (RMSE = 63.40 cycles on the official test

set), consistent with the H1 hypothesis of checkpoint-loading at identical parameter configurations. Both results are therefore reported as a single data point. The anomaly is attributed to the use of identical fixed literature-standard hyperparameters and a potential checkpoint-loading configuration error; the investigation remains open.

Reference: Phase 4/5 Technical Analysis, Section 4.6

6.4 Official Test Set Results

Figure 2 — Model Performance Comparison: Official Test Set, 707 Engines (Iron Wall Protocol)

Official Test Set — 707 Engines (Iron Wall Protocol)

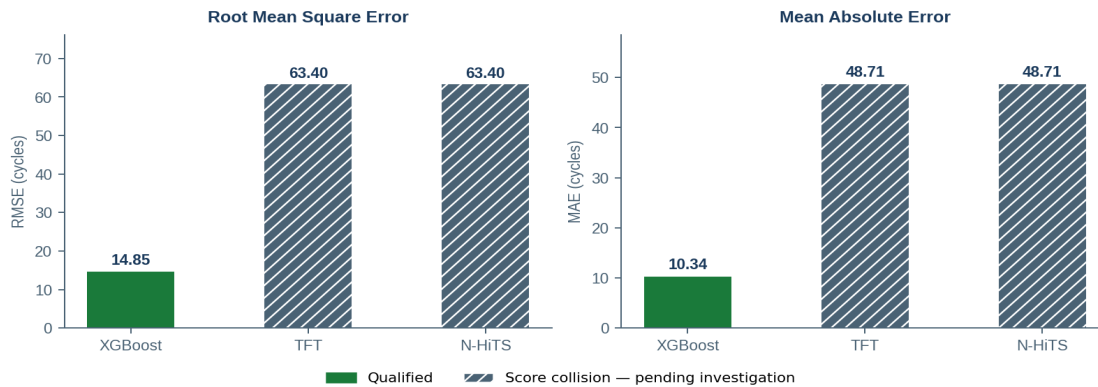


Table 2 — FusionCore v_0 Official Test Set Results Summary (707 Engines, Iron Wall Protocol)

Model	RMSE (cycles)	MAE (cycles)	NASA Score	Critical Recall	Status
XGBoost	14.85	10.34	4,336.3	0.9363	Qualified
TFT	63.40	48.71	1,007,093	0.0446	Score collision — pending HPO
N-HiTS	63.40	48.71	1,007,093	0.0446	Score collision — pending HPO
DeepAR	—	—	—	—	Disqualified

The XGBoost Critical-band recall of 0.9363 — correctly classifying 147 of 157 Critical engines with zero Healthy-to-Critical misclassifications — is the primary safety-critical result of the programme. Under operational doctrine, a Healthy-to-Critical misclassification (predicting adequate RUL for an engine that is actually in the terminal degradation band) represents the most consequential error class. XGBoost produces none of these on the official test set.

6.5 Gate Outcomes

Phase 5 Gate Verification: PASS — 6 gates, all architectures

Reference: Phase 4/5 Technical Analysis, Sections 3–6.

7. Financial & Operational Impact

The financial analysis translates FusionCore v_0 classification performance directly into fleet-scale economic terms. The model operates as a condition-based maintenance decision tool: engines correctly classified as Critical are flagged for planned intervention, avoiding AOG events; engines

correctly classified as Healthy are cleared to continue operations, avoiding unnecessary over-maintenance. The net saving arises from the balance of these two effects. For every ten aircraft an operator flies, the pipeline is projected to save in the region of \$250,000 per year simply by replacing reactive engine removals with planned, scheduled interventions — fewer groundings, fewer disrupted flights, lower recovery costs.

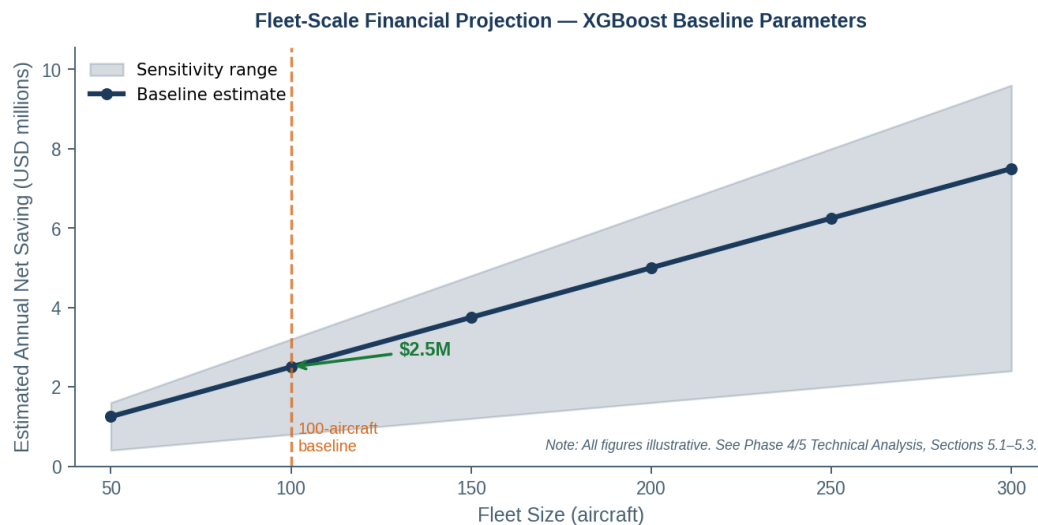
Table 3 — Financial Model Baseline Parameters and Sensitivity Range

Parameter	Baseline Value	Sensitivity Range
AOG cost per event	\$100,000 / day	\$80,000 – \$200,000
UER rate	1% of fleet per annum	0.5% – 2.0%
Over-maintenance cost (avoided)	\$50,000 / event	\$30,000 – \$80,000
Annual net saving (100 aircraft)	~\$2,500,000	~\$800,000 – ~\$2,500,000

The full fleet-scale projection is presented in Figure 3. The sensitivity band reflects the published range of AOG costs from Oliver Wyman MRO Survey (2023) and ACMG (2022) operational cost benchmarks. All figures are illustrative and constructed from transparent assumptions; the full derivation can be found from the reference location.

Reference: Phase 4/5 Technical Analysis, Section 5

Figure 3 — Fleet-Scale Net Annual Savings by Fleet Size (XGBoost Baseline Parameters)



The financial case rests on the Critical-band recall result. A model with recall 0.9363 prevents 93.6% of avoidable unscheduled removals within its operating window. The residual 6.4% miss rate is accounted for in the sensitivity analysis. The zero Healthy-to-Critical misclassification rate is the operational safety assurance: the model does not clear an engine that is actually in terminal degradation. Put simply — on the 707 engines evaluated, the model never told an operator that a failing engine was safe to keep flying.

8. Conclusions & Forward Look

FusionCore v_0 has achieved its stated objectives. The five-phase physics-grounded pipeline demonstrates that a rigorously engineered feature set, derived from first principles of turbofan thermodynamics and degradation mechanics, delivers strong RUL estimation performance on the

NASA C-MAPSS benchmark. XGBoost achieves RMSE = 14.85 cycles and NASA Score = 4,336.3 on 707 official test engines under the Iron Wall Protocol — results that are fully traceable to the physics established in Phase 1 through the feature engineering of Phase 3. All 22 programmatic gate conditions across the five phases pass without exception.

The benchmarking exercise identifies a clear performance gap between the XGBoost baseline (RMSE = 14.85) and the SOTA neural network candidates (RMSE = 63.40). This gap is attributed to the inability of TFT and N-HiTS — as implemented under fixed capacity-scaled hyperparameters without Optuna HPO — to exploit the multi-scale temporal structure of the 91-feature manifold. The score collision anomaly (TFT and N-HiTS producing identical metrics) remains an open finding; resolving it is not within the scope of FusionCore v_1 , which is focused exclusively on PiNet development.

8.1 FusionCore v_1 — PiNet

FusionCore v_1 will introduce PiNet, a hybrid physics-informed network architecture that combines the physics manifold established in Phases 1–3 with a temporal neural encoder. PiNet addresses the core limitation identified in Phase 4: XGBoost exploits instantaneous physics features effectively but cannot model the temporal accumulation of degradation that the survival analysis framework in Phase 3 demonstrates is present in the data. FusionCore v_1 inherits the Phase 3 feature matrices and Phase 2 data splits directly, enabling a controlled comparison against the Phase 4/5 XGBoost baseline on identical data.

Reference: Phase 4/5 Technical Analysis, Section 7.4

Appendix A — Notebook Register

The following table provides the complete register of all Jupyter notebooks in the FusionCore v_0 series, with their titles and purposes. All notebooks are structured to execute sequentially and reproducibly; no cell produces output that is not captured by a downstream programmatic gate.

Table A1 — FusionCore v_0 Notebook Register (Notebooks 01–05e)

Notebook	Title	Purpose
01	EDA & Physics Grounding	Corpus statistics, dead sensor identification, ACF/PACF serial correlation analysis, lifecycle trajectory visualisation, Weibull lifespan modelling, and Phase 1 gate verification.
02	Zero-Leakage Normalisation & Dataset Unification	Iron Wall Protocol partitioning (GroupShuffleSplit), K-Means regime clustering for FD002/FD004, per-regime z-score normalisation, SPC P-F interval analysis, FD00u unified dataset construction, and Phase 2 gate verification.
03	Physics-Aware Feature Engineering	Resolution of 13 Phase 1/2 gaps, construction of the 91-feature physics matrix (kinematic expansion, virtual sensors, Miner's Rule fatigue indices), survival analysis (Kaplan-Meier, Cox PH), UWL threshold recalibration, and Phase 3 gate verification.
04a	XGBoost Baseline Training & SHAP	XGBoost training with early stopping, SHAP TreeExplainer feature attribution, validation performance evaluation.
04b	DeepAR Evaluation & Disqualification	Programmatic architectural audit of DeepAR; confirmation of two disqualifying barriers (no exogenous features, look-ahead contamination).
04c	TFT Training	Temporal Fusion Transformer training with capacity-scaled hyperparameters and NeuralForecast; Phase 4 validation results.
04d	N-HiTS Training (PatchTST Substitution)	N-HiTS training as PatchTST substitute; Phase 4 validation results; score collision diagnostic.
04e	Phase 4 Financial Analysis	Fleet-scale financial projection under Phase 4 validation metrics; baseline and sensitivity analysis.
05a	Test Set Preparation (Iron Wall Protocol)	Official test set ingestion under Iron Wall constraints; application of frozen Phase 2 normalisation and Phase 3 features.
05b	Four-Quadrant Official Evaluation	Quadrant A (RMSE/MAE), Quadrant B (NASA score), Quadrant C (operational classification), Quadrant D (explainability) evaluation on 707 official test engines.
05e	Phase 5 Gate Verification & Financial Projection	Programmatic Phase 5 gate assertion across all six conditions; updated financial projection under official test metrics.

Appendix B — Document Register

The following table provides the complete register of the four FusionCore v_0 Technical Analysis documents. Each document is the primary reference for its respective phase or phases and should be consulted for full methodology, gate verification records, and supporting analysis.

Table B1 — FusionCore v_0 Document Register

Document	Phases	Contents & Purpose
Phase 1 Technical Analysis	Phase 1	Physics audit of the NASA C-MAPSS corpus. Covers corpus statistics (160,359 observations, 709 engines), dead sensor identification (7 sensors confirmed), ACF/PACF serial correlation analysis, KDE validation, Weibull lifespan modelling, lifecycle trajectory and cumulative damage analysis. Establishes the physical credibility of the dataset as the foundation for all downstream modelling. Contains 4 programmatic gate verification records.
Phase 2 Technical Analysis	Phase 2	Zero-leakage preprocessing pipeline. Covers Iron Wall Protocol partitioning, K-Means operating regime identification (6 regimes for FD002/FD004), per-regime z-score normalisation with sigma guard, Shewhart SPC P-F interval analysis (overall CV = 1.052, advisory), dataset unification (FD00u). Establishes the leakage-free normalised feature space for Phase 3. Contains 6 programmatic gate verification records and the Phase 1 gap resolution audit.
Phase 3 Technical Analysis	Phase 3	Physics-aware feature engineering and gap resolution. Covers resolution of 13 Phase 1/2 gaps, construction of the 91-feature matrix (24 core, 51 kinematic, 3 virtual sensors, 3 fatigue indices, gap resolution features), Kaplan-Meier and Cox Proportional Hazards survival analysis, per-fault-mode-family UWL threshold recalibration, and business impact analysis. Contains 6 programmatic gate verification records and 9 formally transferred items to Phase 4.
Phase 4/5 Technical Analysis	Phases 4 & 5	SOTA benchmarking, forensic audit, and official evaluation. Covers transfer item resolution (9 items), XGBoost training and SHAP attribution, forensic null and perturbation tests, DeepAR disqualification, TFT and N-HiTS training, four-quadrant official evaluation on 707 test engines (RMSE = 14.85, NASA Score = 4,336.3, Critical Recall = 0.9363), score collision diagnostic, fleet-scale financial analysis (~\$2.5M annual saving at baseline), and PiNet FusionCore v1 design brief. Contains 6 Phase 5 programmatic gate verification records.